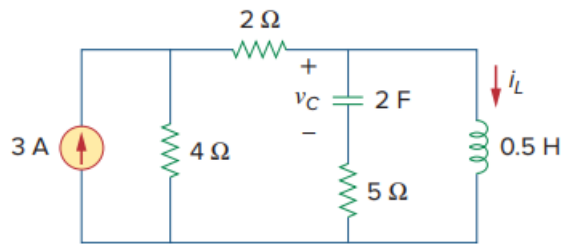


Problem

Find v_C , i_L , and the energy stored in the capacitor and inductor in the circuit of Fig. 6.69 under dc conditions.

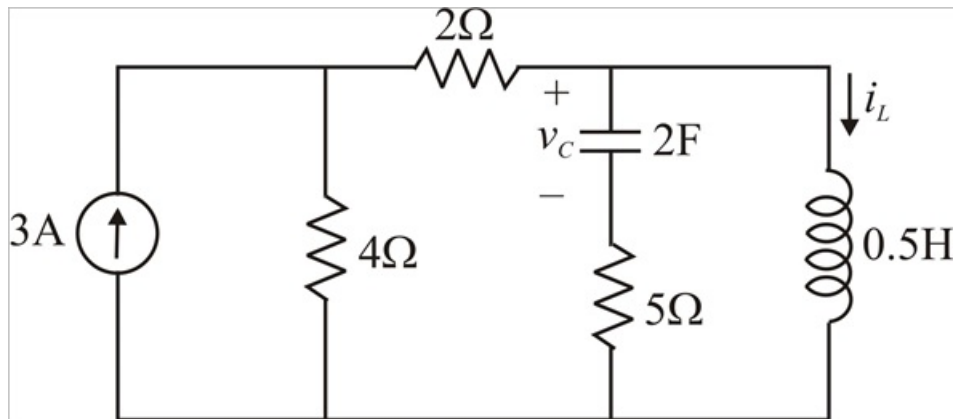
Fig. 6.69:



Step-by-step solution

Step 1 of 5

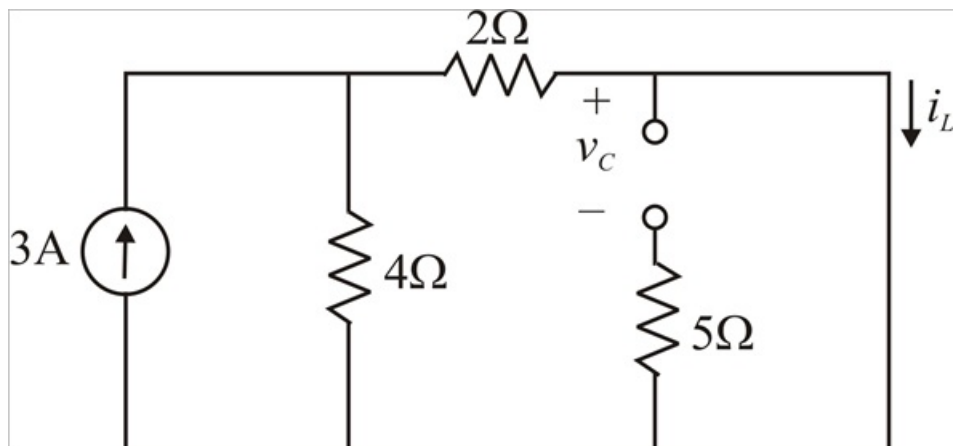
Given circuit diagram



Step 2 of 5

Under dc conditions capacitor acts like an open circuit and inductor acts like short circuit

Now above circuit is modified as below



Step 3 of 5

By using current division rule we can write

$$i_L = 3 \left(\frac{4}{4+2} \right)$$

$$i_L = 2\text{A}$$

Current flowing through 4Ω resistor is

$$i_{4\Omega} = 3 - i_L$$

$$i_{4\Omega} = 3 - 2$$

$$i_{4\Omega} = 1\text{A}$$

Step 4 of 5

By applying KVL to the middle loop we can write

$$-(i_{4\Omega})4 + (i_L)2 + v_C + (0)5 = 0$$

$$-(1)4 + (2)2 + v_C + 0 = 0$$

$$v_C = 0\text{V}$$

[Comments \(1\)](#)



Anonymous

Why is the voltage across the 4(Ohm) resistor the only negative? Shouldn't all resistor voltages be drops (and therefore have the same sign) regardless of which direction you go around the loop?

Step 5 of 5

Energy stored in the capacitor is

$$W_C = \frac{1}{2} C (v_C)^2$$

$$W_C = \frac{1}{2} (2)(0)^2$$

$$W_C = 0\text{J}$$

Energy stored in the inductor is

$$W_L = \frac{1}{2} L (i_L)^2$$

$$W_L = \frac{1}{2} (0.5)(2)^2$$

$$W_L = 1\text{J}$$